Partial differential equation (PDE)

Def^n: A partial differential equation is one which contains more than one independent variables. This type of eq^n plays a very important role in sciences, Economics and engineering.

Let a function is of the form

$$z = f(x, y) \quad \text{——— (1)}$$

Partial derivatives of z w.r.t. x and y are denoted by p and q respectively, so that

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}.$$

Also,

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial p}{\partial x}, \quad s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}, \quad t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial q}{\partial y}$$

* Order and degree of PDE are same as ODE.

Classification of solutions:

1. Complete solution
2. Particular solution
3. Singular solution
4. General solution.

1. Complete solution:

PDE : $3px^2y^2 + 2qy^3x^3 + pq = qx^2y^2z$ (1)

Solution: $z = ax^3 + by^3 + ab$ (2) which is a complete solution, since it contains two arbitrary constants a and b as same number of independent variables.

2. Particular integral (solution):

which has no any arbitrary constants.

Ex: A solution $z = x^3 + y^3 + 1$ is a particular integral.

3. Singular solution:

Differential (2) w.r.t. a and b, we get

$$0 = x^3 + b$$

$$\Rightarrow b = -x^3$$

and $$0 = y^3 + a$$

$$\Rightarrow a = -y^3$$

So (2), becomes, $z = -x^3y^3 - x^3y^3 + x^3y^3$

$z + x^3y^3 = 0$ is the

Singular solution.

Singular solution is a solution of differential equation which is independent of arbitrary constants and which is obtained from the complete solution by eliminating the arbitrary constants

General solution: General solution does not contain any arbitrary constant but is different from singular solution. in Eq^n (2), making b is a function of a, say $b = a$, we get

$$z = a(x^3 + y^3 + a) \quad \text{--- (3)}$$

Diff. w.r.t. a,

$$0 = x^3 + y^3 + 2a \quad \therefore a = -\frac{1}{2}(x^3 + y^3)$$

$\therefore$ (3) $\Rightarrow$ $z = -\frac{1}{2}(x^3 + y^3)\left[x^3 + y^3 - \frac{1}{2}(x^3 + y^3)\right]$

$\therefore\ z = -\frac{1}{2}(x^3+y^3)\,\frac{x^3+y^3}{2}$

$\therefore\ \boxed{4z + (x^3+y^3)^2 = 0}$ → which is a general solution and different from the singular solution.

Linear PDE of 1st order

* If the degree of p and q is one, it is called a linear PDE of order one.

Thus

$$\boxed{px + qy = z}$$

On the other hand, if the degrees of p and q are higher than the first order then PDE is called non-linear.

For Example:

$$\boxed{p^2x^2 + pqxy + q^2y^2 = z^2}$$

<u>Lagrange's linear PDE</u> : A linear PDE of the form $\boxed{pP+qQ=R}$ (1) where P, Q, R are functions of x, y, z, respectively, is ~~called~~ known as Lagrange's linear PDE.

To obtain the solution of the Lagrange's PDE, we have the following rule:

$$\boxed{\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}} \quad (2)$$

which is known as Lagrange's auxiliary eqn

* find the two independent soln of (2) in the form

$$u(x,y,z)=a$$
$$v(x,y,z)=b$$

General soln of (1) can be written in the following

$$\phi(u,v)=0, \quad u=\phi(a), \quad v=\phi(b)$$

* If the PDE is of the form

$$P\frac{\partial z}{\partial x}+Q\frac{\partial z}{\partial y}+R\frac{\partial z}{\partial t}=S \quad (3)$$

then the Lagrange's A.E. ~~are~~ are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{dz}{S} \quad \text{--- (4)}$$

so the G.S. of (3) is

$\phi(u, v, w) = 0$, where u, v, w are three independent soln of (4).

<u>Ex (1)</u> Solve $yzp + zxq = xy$

<u>Ans</u>: A.E. are $\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}$

From first two

$$\frac{dx}{y} = \frac{dy}{x}$$

$$\Rightarrow \int x\,dx = \int y\,dy$$

$$\Rightarrow \frac{x^2}{2} = \frac{y^2}{2} + c_1$$

$$\Rightarrow \boxed{x^2 - y^2 = 2c = c_1}$$

Similarly from the last two

$$\frac{dy}{x} = \frac{dz}{y}$$

$$\therefore \boxed{y^2 - z^2 = c_2}$$

Hence the general solution is

$\phi(x^2-y^2, y^2-z^2)=0$ where ϕ is any arbitrary function.

Ex: ~~Solve~~ ~~$(y+z)p+(z+x)q=xy$~~ $(x+2y^2)dp - xy\,q = xz$

Solve $\dfrac{dx}{x^2+2y^2} = \dfrac{dy}{-xy} = \dfrac{dz}{xz}$

Ans: Taking the last two fractions

$$\int \frac{dy}{-y} = \int \frac{dz}{z} \Rightarrow \ln y + \ln z = \ln c_1$$

$$\Rightarrow yz = c_1$$

Again taking first two fractions

$$\frac{dx}{x^2+2y^2} = \frac{dy}{-xy}$$

$$\Rightarrow \frac{dx}{dy} = \frac{x^2+2y^2}{-xy} \Rightarrow x\frac{dx}{dy} = -\frac{x^2}{y} - 2y$$

$$\Rightarrow x\frac{dx}{dy} + \frac{x^2}{y} = -2y$$

Let $x^2 = u$

$$\Rightarrow \frac{1}{2}\frac{du}{dy} + \frac{u}{y} = -2y \qquad 2x\frac{dx}{dy} = \frac{du}{dy}$$

$$\boxed{\frac{du}{dy} + \frac{2u}{y} = -4y} \qquad I.F = e^{\int \frac{2}{y}dy} = y^2$$

$\Rightarrow$

$$\frac{d}{dy}(y^2 \cdot u) = -y^2 \cdot 4y$$

$$\int d(y^2 u) = -\int 4y^3\, dy$$

$$y^2 u = -\cancel{4}\frac{y^4}{\cancel{4}} + c_2$$

$$y^2 x^2 = -y^4 + c_2$$

$$\Rightarrow x^2y^2 + y^4 = c_2$$

$$\therefore \boxed{\phi(yz,\ x^2y^2 + y^4) = 0}$$

Exercise: ① $\frac{dx}{y^2} = \frac{dy}{x^2} = \frac{dz}{x^2y^2z^2}$

An: $\phi\left(x^3 - y^3,\ x^3 + \frac{3}{z}\right) = 0$

② $xp + yq = z$ An: $\phi(x/z,\ y/z) = 0$

③ $(y^2 z/x)\,p + xzq = y^2$

A: $\phi(x^2 - y^3,\ x^2 - z^2) = 0$.

④ $p\tan x + q\tan y = \tan z$

An: $\frac{\sin x}{\sin y} = \phi(\sin y/\sin z)$.

⑤ $y^2 p - xyq = x(z - 2y)$

An: $\phi(x^2 - y^2,\ zy - y^2) = 0$.

Home tasks:

(1) $p+3q=5z+\tan(y-3x)$

Ans: $5x-\ln[5z+\tan(y-3x)]=\phi(y-3x)$

(2) $z(z^2+xy)(px-qy)=x^4$

Ans: $\phi(xy,\ x^4-y^4-2xyz^2)=0.$

(3) $xyp+y^2q=2xy-2x^2$

Ans: $x-\ln\left[z-2\left(\frac{x^2}{y^2}\right)\right]=\phi(x/y)$

(4) $xzp+yzq=xy.$

Ans: $\phi(xy-z^2,\ x/y)=0.$

(5) $px(z-2y^2)=(z-qy)(z-y^2-2x^3)$

Ans: Given $x(z-2y^2)p+y(z-y^2-2x^3)q=z(z-y^2-2x^3)$ — (1)

The Lagrange's auxiliary equations of (1) is

$$\frac{dx}{x(z-2y^2)}=\frac{dy}{y(z-y^2-2x^3)}=\frac{dz}{z(z-y^2-2x^3)}$$

Taking last two fraction we get

$$\frac{dy}{y\cancel{(z-y^2-2x^3)}}=\frac{dz}{z\cancel{(z-y^2-2x^3)}}$$

Linear PDE : Method of multipliers

By a proper choice of multipliers l, m, n which are not necessarily constants, we may write

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}$$

such that $lP + mQ + nR = 0$. Then

$$l\,dx + m\,dy + n\,dz = 0$$ can be

solved for a solution

$$\phi(x, y, z) = c.$$

Ex-3 (3)

$$\Rightarrow \int \frac{dy}{y} = \int \frac{dz}{z} \Rightarrow \ln z - \ln y = \ln c_1$$

$$\boxed{z/y = c_1}$$

$$\Rightarrow \cancel{\ln y - \ln z = \ln c_1} \Rightarrow$$

$$\Rightarrow \cancel{\ln (y/z) = \ln c_1} \qquad \Rightarrow z = y c_1$$

$$\cancel{(y/z) = c_1} \Rightarrow \boxed{\cancel{y = z c_1}}$$

Now,

$$\frac{dx}{x(z-2y^2)} = \frac{dy}{y(z-y^2-2x^3)}$$

$$\Rightarrow \frac{dx}{x(y c_1 - 2y^2)} = \frac{dy}{y(y c_1 - y^2 - 2x^3)}$$

$$\Rightarrow (y c_1 - y^2 - 2x^3)\, dx = \frac{x\cancel{y}(c_1 - 2y)}{\cancel{y}}\, dy$$

$$\Rightarrow (y c_1 - y^2 - 2x^3)\, dx + x(2y - c_1)\, dy = 0 \quad (2)$$

$$\frac{\partial M}{\partial y} = c_1 - 2y, \quad \frac{\partial N}{\partial x} = 2y - c_1$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{2(2y-c_1)} \times 2(c_1 - 2y) = -\frac{2}{x} = f(x)$$

$$I.F = e^{-\int (2/x)\, dx} = e^{-2\ln x}$$

$$= e^{\ln x^{-2}}$$

$$= \frac{1}{x^2}$$

Ex-4

(2) N.I.F =)

$$\int_{y=\text{const}} \left(\frac{c_1 y}{x^2} - \frac{y^2}{x^2} - 2x\right) dx + \int x^{-1}(2y - c_1)\, dy = 0$$

$$\Rightarrow (c_1 y - y^2)(-1/x) - \frac{2x^2}{2} = c_2$$

$$\Rightarrow \frac{(y^2 - c_1 y)}{x} - x^2 = c_2$$

$$= \frac{y^2 - c_1 y - x^3}{x} = c_2$$

Hence the G.S. is of the form

$$\boxed{\frac{y^2 - c_1 y - x^3}{x} = \phi(z/y)}$$

CW: Solve $x^2 p + y^2 p = z^2$

A.E. $\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{z^2}$

$$\Rightarrow \int \frac{dx}{x^2} = \int \frac{dy}{y^2}$$

$$\Rightarrow -\frac{1}{x} - \frac{1}{y} = c_1$$

$$\int \frac{dy}{y^2} = \int \frac{dz}{z^2}$$

$$\Rightarrow -\frac{1}{y} - \frac{1}{z} = c_2$$

$$\phi\left(-\frac{1}{x} - \frac{1}{y}, \frac{1}{y} - \frac{1}{z}\right) = 0$$

CW

$p + q = \sin x$

A.E. $\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{\sin x}$

$$\Rightarrow \frac{dx}{1} = \frac{dy}{1}$$

$$x - y = c_1$$

$$\frac{dx}{1} = \frac{dz}{\sin x}$$

$$\Rightarrow \int \sin x\, dx = \int dz$$

$$\Rightarrow -\cos x - z = c_0$$

$$\Rightarrow z + \cos x = c_2$$

G.S.

$$\phi(x - y, z + \cos x) = 0$$